



Artificial Intelligence and Machine Learning

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AI Landscape

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Categorical data – **Categorical** variables represent types of **data** which may be divided into groups.

It can be Ordinal or Nominal

Ordinal features – categorical values that can be sorted or ordered.

Eg: T-shirt Size - $XL > L > M$

Nominal – No order in the categorical values

Eg: T-Shirt color

Blue is not greater or lesser than yellow

Treating Ordinal Values - Mapping

```
size_mapping = {'XL':3, 'L':2, 'M':1}
```

```
df['Size']=df['Size'].map(size_mapping)
```

Size		Size
XL		3
L		2
M		1
L		2
XL		3

Treating Nominal - Encoding Class labels

1. Label encoder

```
from sklearn.preprocessing import LabelEncoder  
le = LabelEncoder()  
y = le.fit_transform(df['classlabel'].values)
```

Size		Size
XL		0
L		1
M		2
L		1
XL		0

To inverse to the actual values

```
actual=le.inverse_transform(y)
```

Size		Size
0		XL
1		L
2		M
1		L
0		XL

2. One hot encoding

Creating a new dummy variable for each unique value in the nominal feature

```
from sklearn.preprocessing import OneHotEncoder  
ohe = OneHotEncoder()  
ohe.fit_transform(X).toarray()
```

Colour
Red
Blue
Green
Red
Green



0	1	2
1	0	0
0	1	0
0	0	1
1	0	0
0	0	1

3. Dummies

`pd.get_dummies(X)`

Colour
Red
Blue
Green
Red
Green



Red	Blue	Green
1	0	0
0	1	0
0	0	1
1	0	0
0	0	1



THANK YOU

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